DEFENSYS: A CROSS-PLATFORM CAPSTONE AND PIT MANAGEMENT SYSTEM WITH SECURE ARCHIVING AND CURRICULUM ANALYTICS

A Capstone Project and Research Study

Presented to the

Department of Information Technology
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In Partial Fulfillment of the Requirements for
Capstone Project and Research Study Leading to the Degree in
Bachelor of Science in Information Technology

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

Capstone and Project Integration Technology (PIT) courses serve as the primary performance benchmarks in Information Technology education. However, managing these assessments currently involves highly fragmented and inefficient workflows. During panel defenses, educators rely on a decentralized mixture of paper-based rubrics, individual laptops, and spreadsheet calculations (Trifts & Kohl, 2023). This disconnected approach increases the risk of human computation errors, particularly when calculating complex weighted formulas and consolidating raw instructor points.

Beyond grading inefficiencies, the existing method for archiving project deliverables introduces significant logistical and security vulnerabilities. The department's reliance on physical storage media, compounded by a lack of standardized cloud allocations, creates retrieval bottlenecks during critical evaluations such as ISO accreditation (Díaz et al., 2015). Furthermore, without strict access controls, student intellectual property and system source codes remain vulnerable to unauthorized extraction and plagiarism (Exline, 2016; Hertenstein, 2014). Finally, the current infrastructure lacks the analytical tools necessary to track the emerging technologies students utilize in their projects.

Consequently, curriculum development depends heavily on observational estimates rather than empirical data (Maglapuz & Lacatan, 2023). To resolve these systemic gaps, this study proposes the development of DefenSYS: A Cross-Platform Capstone and PIT Management System with Secure Archiving and Curriculum Analytics.

1.2 Statement of the Problem

This research addresses the lack of a unified system for evaluating and archiving Capstone and Project Integration Technology (PIT) courses within the IT department. Specifically, the study aims to design and develop DefenSYS to answer the following research questions:

1. How can a hybrid Web and Mobile Evaluation Engine accommodate dual grading workflows—computing weighted Capstone grades and aggregating raw PIT points—to eliminate the human computation errors associated with decentralized spreadsheets?
2. How can a file validation gatekeeper automate the archiving process by strictly enforcing departmental naming conventions, thereby resolving labeling and retrieval bottlenecks during ISO audits?
3. How can Role-Based Visibility Filtering protect intellectual property by restricting general student access strictly to stage-specific deliverables (approved PIT papers, proposals, and media summaries), while securing

full-length manuscripts in a read-only, dynamically watermarked digital vault for authorized personnel?

4. How can Machine Learning algorithms (TF-IDF and Naive Bayes) classify document themes to power a Decision Support System (DSS), shifting curriculum development from observational guesswork to the empirical tracking of student technology trends?

1.3 Statement of the Objectives

1.3.1 General Objective

The general objective of this study is to design, develop, and evaluate DefenSYS: a Cross-Platform Capstone and PIT Management System featuring Secure Archiving and Curriculum Analytics, aimed at optimizing the evaluation workflow and securing the academic repository of the College of Information Technology.

1.3.2 Specific Objectives

The study aims to fulfill the following specific objectives:

1. **To design and develop a hybrid Web and Mobile Evaluation Engine** capable of supporting dual grading workflows by computing weighted Capstone grades and aggregating raw PIT points, while integrating a database-level Grade Lock mechanism to ensure the integrity and immutability of submitted scores.

2. **To develop an automated file validation and archiving pipeline** that uses Regular Expression validation to enforce departmental naming conventions and automatically routes approved documents into their proper repository directories.
3. **To construct a secure digital repository with Role-Based Visibility Filtering** that protects student intellectual property by restricting access to full-length manuscripts and allowing general student access only to authorized stage-specific deliverables.
4. **To integrate a Machine Learning-based Curriculum Analytics module** using TF-IDF and Naive Bayes algorithms to classify document themes and support a Decision Support System for tracking student technology trends.

1.4 Significance of the Study

The successful implementation of DefenSYS will significantly benefit the following stakeholders:

- **Department Chairman:** DefenSYS provides a centralized platform to manage academic terms and streamline departmental operations. It delegates routine workflows—such as rubric configuration and team formation—to PIT Leads, while the Machine Learning classifier automates document categorization to maintain an ISO-ready repository. Additionally, the Decision Support System (DSS) equips administrators with empirical data on technological trends for strategic curriculum planning.

- **Faculty Members (Advisers, PIT Leads, and Panelists):** The system eliminates the reliance on paper rubrics and manual spreadsheet computations. Through the mobile Evaluation Engine, Panelists can efficiently execute dual workflows in real-time: computing weighted Capstone grades and aggregating raw PIT scores. Advisers and PIT Leads utilize dedicated web dashboards to securely submit grades and route approved deliverables directly into the automated Validation and AI Pipeline.
- **Temporary Guest Panelists:** The system streamlines external evaluator participation by provisioning dynamic, temporary access codes. This allows guest panelists to securely authenticate and utilize the mobile grading interface without the administrative overhead of permanent account registration.
- **Repository Assistants:** By utilizing a restricted, write-only dashboard, designated utility personnel and student clerks can safely assist faculty in uploading approved deliverables to the file validation gatekeeper. This securely distributes the administrative workload without exposing sensitive evaluation or analytics tools.

- **IT Students:** Students gain a dedicated portal to monitor their team profiles and evaluation statuses. DefenSYS actively protects their intellectual property from AI plagiarism by utilizing Role-Based Visibility Filtering. By restricting digital access solely to authorized, stage-specific deliverables (e.g., approved PIT documents, early proposals, and summarized media), the system enforces the department's mandate for utilizing the physical library to read full-length manuscripts.
- **Future Researchers:** The platform will serve as a highly organized, secure reference repository for future Capstone ideation. Furthermore, its underlying cross-platform architecture and Machine Learning implementation provide a robust technical blueprint for developers engineering future academic and repository management systems.

1.5 Scope and Limitation of the Study

Scope:

DefenSYS is engineered to manage, evaluate, and archive Project Integration Technology (PIT) courses for students from their first year up to the first semester of their third year, and Capstone projects from the second semester of the third year onwards. To align with departmental retention policies, the system accommodates a one-year project extension period by dynamically porting the active status of delayed teams into current semesters, thereby preventing On-the-

Job Training (OJT) clearance until all final deliverables are securely archived. The platform utilizes a hybrid architecture comprising a Web-Based Portal for administration and a Mobile Application for faculty evaluation and student access. It enforces strict Role-Based Access Control (RBAC) across five user tiers: IT Administrators, Faculty, temporary Guest Panelists, Students, and write-only Repository Assistants. The evaluation module automates dual-grading workflows, computing weighted percentage formulas for Capstone defenses while concurrently tracking centralized raw points for PIT evaluations. Furthermore, an integrated Natural Language Processing (NLP) pipeline utilizes TF-IDF and Naive Bayes algorithms to automatically categorize and route approved documents into thematic directories for the Decision Support System (DSS). Finally, the system employs Role-Based Visibility Filtering; while full manuscripts are securely archived for ISO audits and evaluator access, general student viewing is restricted solely to authorized, stage-specific deliverables—such as PIT documents, Capstone proposals, and summarized media—to actively enforce the department's mandate for physical library utilization.

Limitations:

To ensure focused development and system stability, DefenSYS is bound by specific operational limitations. The platform functions exclusively as a defense evaluation and repository management tool; it does not operate as a Learning Management System (LMS) and will not track daily attendance or non-defense academic grading. Due to its cloud-hosted Client-Server architecture, the system requires a stable, active internet connection for real-time grading synchronization and backend AI processing, offering no offline functionality. To protect intellectual property, the system strictly prohibits the digital distribution or general student viewing of full-length Capstone manuscripts and source codes. Additionally, the Machine Learning pipeline is strictly limited to text categorization and trend tracking. It does not evaluate the academic quality of the text, nor does it generate grades, as all grading calculations rely entirely on explicit human inputs. Finally, the system does not execute internet-based plagiarism checks, such as Turnitin, and relies on lightweight probabilistic models rather than computationally expensive Deep Learning algorithms requiring advanced GPUs.

1.6 Definition of Terms

Capstone Project. The final academic requirement for IT students, consisting of a functional software system and a technical manuscript. It spans from the second semester of the third year to the first semester of the fourth year. Delayed teams are granted a maximum one-year extension (to either continue or restart their project) before receiving On-the-Job Training (OJT) clearance.

Decision Support System (DSS). A DefenSYS module that mines repository data and evaluation scores to generate predictive analytics and identify technology trends for curriculum development.

Dynamic Watermarking. A security feature that overlays the viewer's User ID and timestamp onto documents to deter unauthorized reproduction and protect intellectual property.

Grade Lock. A database integrity mechanism that permanently secures evaluation scores and rubrics immediately upon submission, preventing any subsequent alterations.

Guest Panelist. An external evaluator invited by the IT Department. They access the mobile grading interface using dynamically generated temporary codes. While their access expires post-defense, their identity and submitted scores are permanently retained to satisfy ISO audit requirements.

Machine Learning (ML) / Natural Language Processing (NLP). The system's AI pipeline that autonomously extracts and analyzes text from student submissions. It leverages TF-IDF and Naive Bayes algorithms to categorize documents by technological focus and routes them to the DSS.

Naive Bayes Classifier. A probabilistic machine learning algorithm utilized by the system to classify documents into specific technological themes (e.g., machine learning, web development) based on extracted text patterns.

PIT Lead. A faculty member responsible for coordinating evaluation events and managing student team formations for a specific year level.

Project Integration Technology (PIT). Benchmark foundational project development courses taken by IT students from their first to third years.

Rapid Application Development (RAD). The agile software development methodology used to build DefenSYS, characterized by iterative prototyping and continuous user feedback.

Repository Assistant. A restricted user role assigned to utility personnel or student clerks. It grants strictly write-only privileges to upload faculty-approved deliverables into the Validation and AI Pipeline.

TF-IDF (Term Frequency-Inverse Document Frequency). A statistical algorithm employed by the NLP pipeline to evaluate the weight and importance of specific terms within an uploaded document, facilitating accurate thematic extraction.

CHAPTER II

REVIEW OF RELATED LITERATURE AND RELATED SYSTEMS

2.1 Review of Related Literature

The management of capstone projects involves a multifaceted workflow, ranging from live panel evaluation to the final archiving of student manuscripts. Traditionally, these benchmark assessments have relied on fragmented methodologies. Trifts and Kohl (2023) note that the dependence on physical rubrics and scattered spreadsheet applications frequently leads to disjointed evaluations and computational inaccuracies. To resolve these administrative bottlenecks, educational institutions are increasingly transitioning toward integrated learning and digital repository systems (Díaz et al., 2015). However, many existing platforms lack the flexibility to handle varied grading structures. To address this, DefenSYS introduces a hybrid Evaluation Engine capable of concurrently processing complex weighted Capstone computations and Project Integration Technology (PIT) point accumulations.

As digital repositories expand to include undergraduate research, protecting student intellectual property from unauthorized extraction and AI-assisted plagiarism has become a critical priority (Exline, 2016; Hertenstein, 2014). Literature on information security emphasizes that basic cloud storage is inadequate. Modern frameworks require robust access control models, such as Role-Based Access Control (RBAC), to adapt to varying user privileges and mitigate vulnerabilities (Bello et al., 2025; Farhadighalati et al., 2025). DefenSYS applies RBAC alongside dynamic watermarking to secure its read-only digital vault. By digitally embedding the viewer's identity over the document, watermarking serves as a direct deterrent against unauthorized reproduction. This security architecture ensures that while stage-specific deliverables are accessible to students, full manuscripts remain protected, actively enforcing the university's mandate for physical library research.

Before documents can be securely archived, they must be properly standardized. Manual organization of school-related documents is highly prone to human error and creates significant retrieval difficulties (Buctuanon et al., 2021). Regular expressions (Regex) offer a reliable automated solution for input validation and text extraction (Tariq & Rana, 2025). By implementing a Regex gatekeeper, DefenSYS strictly enforces departmental file-naming conventions, automatically rejecting non-compliant uploads to maintain a highly organized, ISO-ready database.

Finally, the categorization of large document volumes requires efficient automated solutions. Recent studies demonstrate the effectiveness of Natural Language Processing (NLP) in text classification. The Term Frequency-Inverse

Document Frequency (TF-IDF) algorithm is widely used to extract key features and measure the relative importance of words within a dataset (Oruç & Senyürek, 2025; Das et al., 2023). When combined with the Naive Bayes classifier—a probabilistic machine learning model praised for its high computational efficiency and capacity to handle high-dimensional data without the resource demands of deep learning (Pajila et al., 2023; Ngo Van et al., 2026)—systems can accurately categorize text (Zhang, 2025). By utilizing TF-IDF and Naive Bayes, DefenSYS automates the thematic routing of approved manuscripts, seamlessly feeding structured data into the Decision Support System (DSS) to track curriculum and technology trends.

2.2 Review of Related Application

This section discusses the various integrated systems and academic management software applications developed to support capstone projects, document repositories, and academic analytics. With the move away from fragmented management systems in higher education institutions, various software applications have been implemented to support centralized evaluation and archiving. The following subsections discuss existing systems, focusing on technology limitations and the gaps addressed by the DefenSYS architecture.

2.2.1 Integrated Platform for Capstone Project Management

Upadhyay and his team developed this system in 2024. The system uses Next.js and Node.js to automate capstone projects in a decentralized environment. The system has an automated system that provides functional dashboards tailored to specific user roles. The system enables users to create capstone projects, submit proposals, and conduct peer reviews. The system lacks a strict Regex file validation gatekeeper. The system does not automatically use Machine Learning (TF-IDF and Naive Bayes) to perform categorization and directory routing for the uploaded repository.

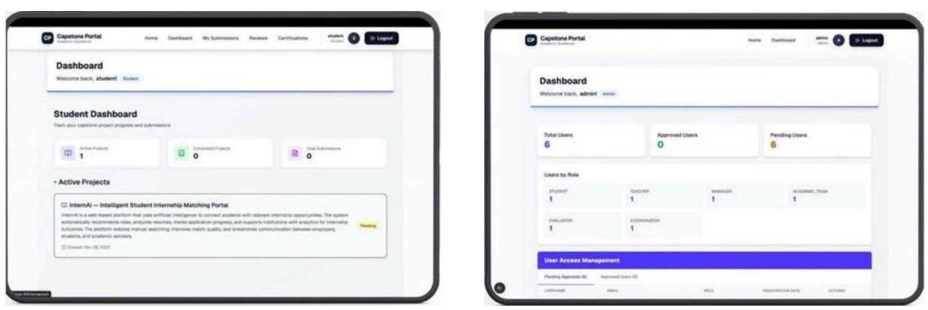

Figure 1. *User Interface of the Integrated Platform for Capstone Project Management*

Figure 1 shows a decentralized platform developed with Next.js and Node.js. The manuscript covers role-tailored dashboards for project proposals and peer reviews but also highlights the missing strict Regex file validation gatekeeper and the lack of Machine Learning for automated document categorization.

2.2.2 Manuscript Archiving Management System

This is a digital archiving management system developed by Perante et al. (2024) using the Rapid Application Development (RAD) approach. This system is strictly focused on the digitalization of traditional paper archives and physical library storage. This digital archiving system has robust search parameters, enabling users to retrieve completed research projects by filtering titles, authors, and advisers. A significant weakness of the digital archiving management system lies in its insufficient safeguards for intellectual property against digital threats and AI-driven plagiarism detection tools. In contrast to DefenSYS, the digital archiving management system does not incorporate a read-only digital vault, which is essential for protecting research projects from plagiarism. Furthermore, the digital archiving management system does not adhere to the physical library mandate, thus failing to limit general viewing access to authorized stage-specific deliverables, including PIT documents, preliminary Capstone proposal works, and summarized media works (Perante et al., 2024).

Figure 2*. User Interface of Manuscript Archiving Management System*

Figure 2 shows a digitalized traditional paper archive system with search parameters to retrieve completed research.

2.2.3 Capstone Project Management System

This is a digital application developed by Yang (2023) using the Waterfall approach. This is a web application developed using Java, Tomcat, and NoSQL Firebase architecture. The digital application features responsive interfaces, dynamic student access, adviser commentary, and role-based access control for team organization. A significant deficiency of this digital application is the absence of a dedicated evaluation engine. Specifically, the application lacks the dynamic grading logic required for dual workflows, preventing the concurrent computation of complex weighted percentages for Capstone defenses while also serving only as a raw point tracker for lower-level PIT evaluations. Furthermore, the digital application lacks the essential database-level Grade Lock feature, which is crucial for preventing grades from being altered after submission.

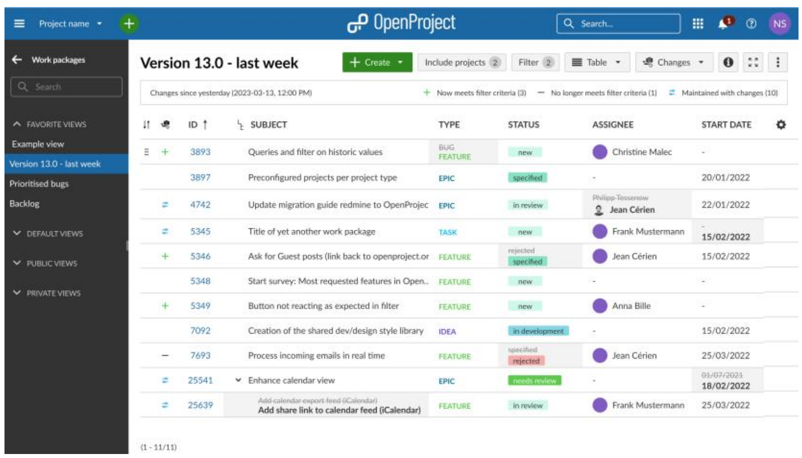

Figure 3. *User Interface of Capstone Project Management System*

Figure 3 shows a Java and NoSQL Firebase based application with responsive interfaces and role-based access for team organization.

2.2.4 Curricular Analytics App

Slim and his team (2022) used advanced data mining and Directed Acyclic Graphs to analyze complex curricula. Their goal was to identify factors that hindered student progress. It incorporates historical data and Monte Carlo simulations to develop data-informed decision-making tools for administrators. Although it supports the hypothesis by demonstrating the need to track bottlenecks in student progress, the application is the greatest limitation, as it is used in a completely isolated manner. The application created by DefenSYS is more advanced, as it includes a Decision Support System (DSS) integrated with the main application to track specific programming languages or technology trends.

Figure 4. *User Interface of Curricular Analytics App*

Figure 4 shows Application Interface using advanced data mining to analyze curriculum to identify bottlenecks in student progress.

2.2.5 Summary of Related Applications

To visually establish the baseline for the proposed research, the next section presents a table summarizing the features and functionalities of related applications in relation to the proposed system.

Table 1. *Comparison of Related Systems*

System/Application	Project & Evaluation Workflow	Secure Archiving & IP Protection	Automated ML Classification	Curriculum Analytics (DSS)
Integrated Platform *(Upadhyay et al., 2024)*	✓	X	X	X
Manuscript Archiving System *(Perante et al., 2024)*	X	✓	X	X
Project Management System *(Yang, 2023)*	✓	✓	X	X
Curricular Analytics App *(Slim et al., 2022)*	X	X	X	✓
DefenSYS (Proposed System)	✓	✓	✓	✓

Table 1 shows a visual comparison of the related systems based on project and evaluation workflow, secure archiving and IP protection, automated ML classification, and curriculum analytics.

2.3 Synthesis and Gap Analysis

A synthesis of pertinent literature reveals that while transitioning to digital evaluation (Bhandari et al., 2024; Upadhyay et al., 2024), digital archiving (Perante et al., 2024), and curriculum modeling (Slim et al., 2022) significantly reduces administrative burdens, current solutions remain highly fragmented. Existing applications typically function as isolated tools—operating either as basic document drives or standalone management trackers (Yang, 2023). Crucially, these systems lack the flexible dual-grading capabilities required to handle varied evaluation structures. Furthermore, they operate as "blind" storage repositories without automated validation gatekeepers, leading to disorganized archives. Most alarmingly, they fail to implement robust access controls or read-only parameters for full-length manuscripts, leaving student intellectual property highly vulnerable to unauthorized extraction and AI-assisted plagiarism.

DefenSYS is engineered to systematically resolve these gaps by unifying evaluation, automated archiving, and curriculum analytics into a cohesive, secure platform. To address fragmented grading, DefenSYS introduces a hybrid Evaluation Engine capable of concurrently processing complex weighted Capstone formulas and raw PIT point totals. To eliminate disorganized storage, the system replaces manual sorting with a strict Regex file-validation gatekeeper paired with a

Natural Language Processing (NLP) pipeline. Utilizing TF-IDF and Naive Bayes algorithms, this pipeline autonomously classifies and routes approved documents into correct thematic directories without human intervention.

Furthermore, DefenSYS mitigates intellectual property vulnerabilities through a read-only digital vault secured by Role-Based Visibility Filtering, database-level "Grade Locks," and dynamic watermarking. By restricting general student access exclusively to stage-specific deliverables (e.g., approved PIT documents and early proposals), the system actively enforces the mandate for physical library utilization while preserving complete manuscripts for authorized evaluators. Finally, DefenSYS bridges the analytical gap by integrating a Decision Support System (DSS) directly into the repository, empowering administrators to empirically track emerging technology trends for continuous curriculum development.

CHAPTER III

METHODOLOGY

3.1 Research Methodology

DefenSYS's development will employ the Rapid Application Development (RAD) methodology. This agile framework was selected to maximize adaptability and maintain strict adherence to the IT department's operational protocols through iterative prototyping and direct user feedback. As depicted in Figure 5, the RAD model commences with the Requirements Planning phase, where the research team conducts interviews and operational observations with IT Administrators, PIT Leads, Faculty, Students, and Repository Assistants. This initial phase defines critical system rules, specifically mapping the mathematical logic for the dual-grading engine—which processes Capstone weights and PIT raw points—and establishing the access constraints for the read-only digital vault to actively enforce physical library usage.

Following requirements analysis, the User Design phase focuses on architecting the system and designing user interfaces. During this stage, researchers finalize the cross-platform technology stack, utilizing Flutter to deploy both the mobile evaluation application and the WebAssembly (Wasm) administrative dashboard from a single codebase. The backend architecture is established using Django, alongside Python-based libraries for the Natural Language Processing (NLP) pipeline. The team also constructs the relational PostgreSQL data models required to support the database-level "Grade Lock" and Role-Based Access Control (RBAC). Instead of a linear coding approach, the subsequent Rapid Construction phase builds the system through iterative prototyping cycles. Developers construct functional prototypes for key modules, including the hybrid evaluation engine, the Decision Support System (DSS), and the Machine Learning classifier utilizing TF-IDF and Naive Bayes algorithms, allowing for immediate faculty and administrator feedback.

Finally, the Cutover phase encompasses containerized system deployment and rigorous testing procedures, including unit, integration, and User Acceptance Testing (UAT). To ensure the system is fully prepared for ISO audits, researchers will conduct strict accuracy evaluations on the Regex file validation gatekeeper and the AI-driven document classifier. This final phase concludes with the formulation of detailed system documentation and orientation sessions, guaranteeing a seamless official handover to the IT department.

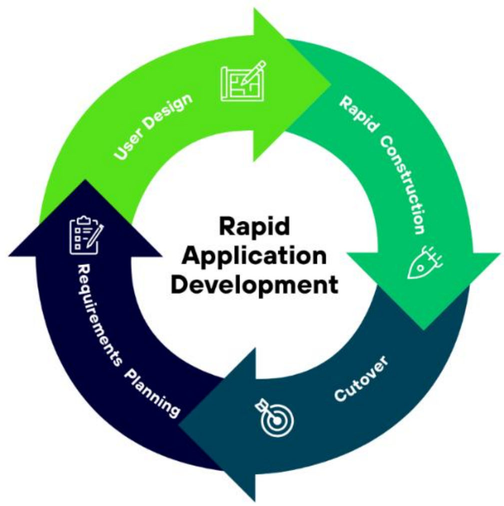

***Figure 5**. Rapid Application Development (RAD) Methodology*

Figure 5 shows the Rapid Application Development with the four iterative phases of Requirements Planning, User Design, Rapid Construction and Cutover.

3.2 Data Gathering Procedure

To ensure that DefenSYS accurately addresses the operational bottlenecks within the Department of Information Technology, the researchers will use a developmental research design with qualitative requirements gathering and quantitative software evaluation. During the requirements planning phase, qualitative data will be gathered through key informant interviews and document/archive analysis to identify the current workflow problems, system requirements, grading procedures, file-naming rules, and access control needs. After the prototype is developed, quantitative data will be gathered through a software evaluation questionnaire based on the ISO 25010 software quality model to assess the system's functional suitability, usability, reliability, and security. The following methods will be utilized:

- **Key Informant Interviews:** Semi-structured interviews will be conducted with primary stakeholders, including IT Administrators, PIT Leads, Capstone Advisers, and Panelists. These interviews aim to map specific workflow inefficiencies, specifically targeting the manual encoding of team data, the fragmentation of grades collected from multiple panelists, and the security vulnerabilities surrounding student intellectual property.
- **Document and Archive Analysis:** Researchers will conduct a comprehensive audit of the department's existing grading rubrics and digital repositories. Acquiring the current rubrics allows developers to program the precise mathematical logic required for the automated calculation engine, differentiating between weighted Capstone formulas and raw PIT point metrics. Furthermore, evaluating past repository submissions will expose current file-naming inconsistencies, which will directly inform the validation rules for the Regex gatekeeper. This archive analysis also

provides the foundational textual dataset necessary to train the NLP pipeline (TF-IDF and Naive Bayes algorithms). Ultimately, the data extracted from these administrative documents will directly dictate the architectural design, database schema, and Role-Based Visibility constraints of the digital vault.

3.2.1 Research Instrument

To systematically execute the data gathering procedures, the researchers will utilize three primary instruments to assess user workflows and define technical system requirements:

- **Semi-Structured Interview Guide:** Adapted from established Requirements Elicitation frameworks (Valacich & George, 2020; Sommerville, 2015), this tool utilizes open-ended questions to interview IT Administrators, Faculty, and Students. It is explicitly designed to pinpoint specific workflow bottlenecks in real-time grading and define critical intellectual property security needs. (A copy of this guide is provided in the Appendices).
- **Documentary Analysis Matrix:** This matrix serves as an audit tool for existing departmental repositories and administrative documents. It is utilized to extract the precise mathematical formulas from current paper

rubrics to program the dual-grading engine (Capstone weights vs. PIT raw points). Furthermore, it extracts historical textual data to train the NLP pipeline for thematic categorization, while defining the archival parameters necessary to enforce Role-Based Visibility Filtering for stage-specific deliverables.

- **Software Evaluation Questionnaire:** Administered following the Rapid Construction phase, this standardized survey is grounded in the ISO 25010 Software Quality model. Purposively selected respondents will utilize a 5-point Likert scale to quantitatively evaluate the DefenSYS prototype's functional suitability, usability, and reliability.

3.2.2 Sampling Method

The researchers will employ Purposive Sampling to identify and select respondents for data gathering. Because DefenSYS is a highly specialized administrative tool, random sampling would fail to capture the precise technical requirements needed for system development. Instead, the study targets 15 key personnel within the Department of Information Technology who possess direct, specialized knowledge of Capstone and PIT workflows.

The respondents are purposefully selected based on the following criteria:

- **One IT Administrator:** Selected to define the exact archival workflows and establish Role-Based Visibility Filtering. This ensures general student access is strictly restricted to stage-specific deliverables (approved PIT documents, Capstone proposals, and summarized media). The administrator will also define security access controls and identify the historical datasets required to train the Decision Support System (DSS).
- **Four Faculty Members (Capstone Advisers, PIT Leads, and Panelists):** Selected to map the mechanics of mobile evaluation and identify logistical bottlenecks during live defense sessions. Crucially, this group will explicitly define the mathematical parameters required for the dual-grading engine (separating weighted Capstone formulas from raw PIT point totals).
- **Ten IT Students (1st to 4th Year):** Purposively chosen from active PIT and Capstone cohorts to define the User Experience (UX) requirements for the read-only digital vault, document submission protocols, and the real-time peer-to-peer evaluation interface.

By using purposive sampling, the researchers selected respondents who have direct experience with Capstone and PIT workflows. Their feedback will help ensure that DefenSYS is designed and evaluated based on the actual needs of its intended users.

3.2.3 Statistical Treatment of Data

To systematically analyze the data gathered during the testing and evaluation of the DefenSYS platform, the researchers will utilize two statistical methods.

1. Weighted Mean (Software Quality Evaluation) To quantitatively assess the platform's functional suitability, usability, and reliability based on the ISO 25010 Software Quality standard, data gathered from the 5-point Likert scale survey will be treated using the weighted mean formula:

$$\bar{x} = \frac{\sum fx}{N}$$

Where:

- $\bar{x}$ = Weighted Mean
- f = *frequency of responses*
- x = weight/value of each response
- N = Total number of respondents

To interpret the gathered data from the Likert scale, the following standard interpretation table will be utilized:

Table 2. *Likert Scale Interpretation Guide*

Range	Mean Score	Qualitative Interpretation
5	4.50 – 5.00	Highly Acceptable (Strongly Agree)
4	3.50 – 4.49	Acceptable (Agree)
3	2.50 – 3.49	Moderately Acceptable (Neutral)
2	1.50 – 2.49	Less Acceptable (Disagree)
1	1.00 – 1.49	Not Acceptable (Strongly Disagree)

Table 2 shows the weighted mean ranges (1.00 - 5.00) and the qualitative descriptions (Not Acceptable to Highly Acceptable) used in the assessment of the functional suitability, usability, and reliability of the system.

2. Confusion Matrix Metrics (Machine Learning Evaluation)

To objectively evaluate the Natural Language Processing (NLP) pipeline (specifically the TF-IDF feature extraction and Naive Bayes classifier), the researchers will utilize a confusion matrix. Because the Regex gatekeeper validates file parameters prior to classification, the AI is evaluated strictly on its thematic categorization accuracy. For this study, the target accuracy for the classification module is set at a minimum threshold of 85%. The 85% accuracy threshold was selected as a practical benchmark to determine whether the classification model is reliable enough for assisting repository organization while still allowing human verification for incorrectly classified documents.

To visualize the model's performance, the predictions will be mapped using the following standard confusion matrix structure:

Table 3. Confusion Matrix Layout

	Predicted Theme (Positive)	Predicted Theme (Negative)
Actual Theme (Positive)	True Positive (TP)	False Negative (FN)
Actual Theme (Negative)	False Positive (FP)	True Negative (TN)

Table 3 shows the evaluation setting for the AI thematic classification accuracy by aligning True Positives (TP), False Negatives (FN), False Positives (FP) and True Negatives (TN).

Using the data from this matrix, the primary metric to validate predictive performance is Accuracy, calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP = True Positives are the correct classification of topics of documents (such as a machine learning paper that should go to the Machine Learning folder).
- TN = True Negatives are the correct classification of non-matches.
- FP = False Positives are the incorrect classification of a document under a particular topic (for example, a Web Development paper that is classified as belonging in the Machine Learning folder).
- FN = False Negatives refer to documents that belong to a specific theme but were incorrectly classified under another theme or not detected as part of that theme.

To ensure that no particular document class is favored, additional measurements are needed which will include Precision ($Precision = \frac{TP}{TP+FP}$) and Recall ($Recall = \frac{TP}{TP+FN}$), calculated from the confusion matrix.

3.3 Description of the Current System (Information Gathering)

Currently, the Department of Information Technology manages its Project Integration Technology (PIT) and Capstone defense procedures through a highly fragmented, manual, and semi-digital workflow. This decentralized infrastructure introduces significant operational bottlenecks across three primary domains: real-time evaluation, document archiving, and intellectual property security.

During live defense sessions, panelists and advisors rely primarily on physical paper rubrics or isolated spreadsheet files. Following the presentations, faculty members must manually consolidate these disparate grading sheets to compute final scores. This manual aggregation is highly susceptible to human computation errors, particularly when instructors must alternate between calculating complex, weighted percentage formulas for Capstone defenses and aggregating raw point totals for PIT evaluations. Furthermore, peer-to-peer evaluations are tracked manually, complicating the accurate assessment of individual student contributions and severely delaying the official grading timeline.

Beyond evaluation inefficiencies, the existing archival workflow for final project deliverables relies entirely on unstructured cloud storage links and local file transfers. Because the infrastructure lacks an automated validation gatekeeper, students frequently submit files with inconsistent naming conventions. Consequently, repository administrators must expend considerable effort at the end of every semester to manually open, verify, rename, and read through documents to categorize them into thematic departmental folders. This lack of an automated classification pipeline creates severe retrieval bottlenecks during critical ISO audits.

Finally, the current infrastructure operates merely as a static file repository, lacking the security and analytical capabilities required of a modern academic system. Without Role-Based Visibility Filtering or read-only access controls, the unrestricted digital distribution of full-length manuscripts exposes student intellectual property to unauthorized extraction and AI-assisted plagiarism. This failure to isolate stage-specific deliverables directly undermines the department's mandate for physical library utilization. Furthermore, the absence of an automated Decision Support System (DSS) prevents IT Administrators from mining repository data, severely impeding their ability to track emerging technological frameworks for prospective curriculum revisions.

3.4 System Architecture and Development

The Cross-Platform Capstone and PIT Management System (DefenSYS) is built upon a contemporary, multi-tiered Client-Server architecture. This design ensures that resource-intensive processes, including grade calculations and machine-learning classification, are securely managed on the backend server while simultaneously offering a swift and responsive interface for end users.

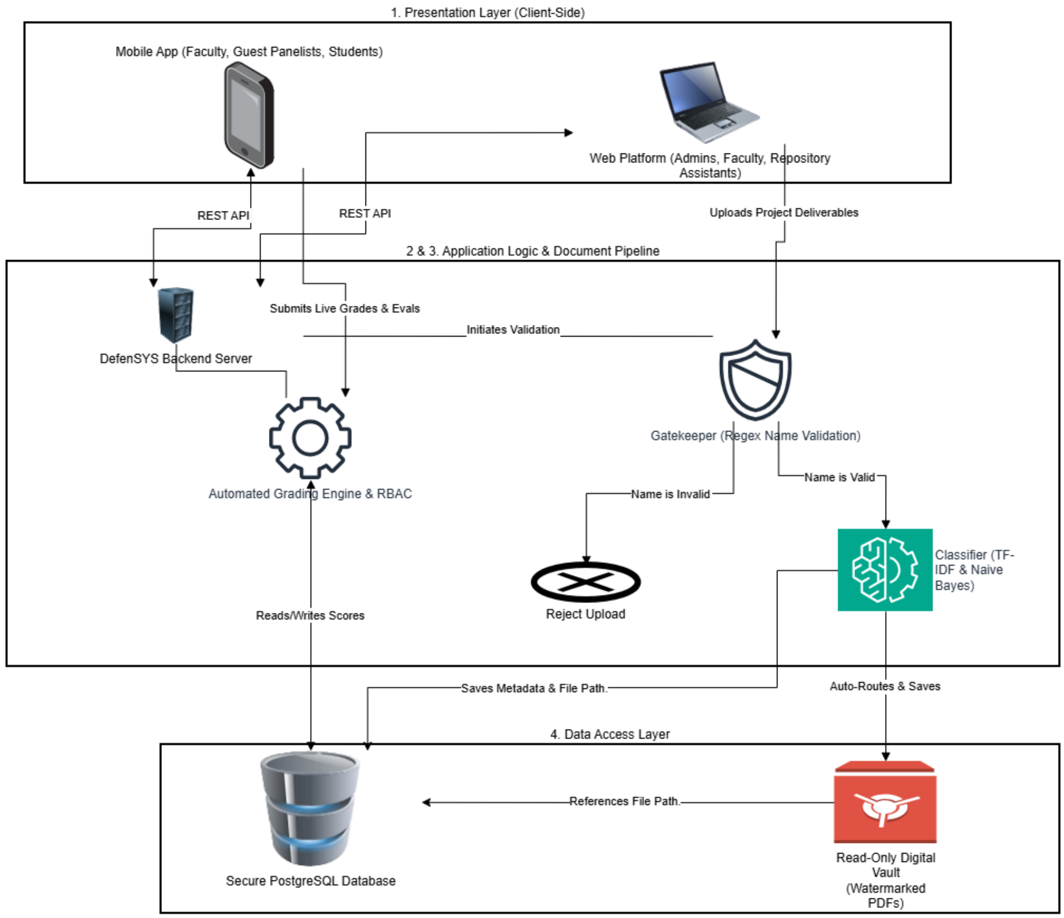

Figure 6. *System Architecture and Development Diagram*

Figure 6 shows the multi-tiered Client-Server architecture of the DefenSYS platform, mapping the seamless integration between the client-side presentation interfaces, the Django-powered application logic, and the secure data access layer.

The architecture is systematically divided into four primary components:

Presentation Layer (Client-Side) Serving as the primary user interface, this layer is built using a unified Flutter codebase and is deployed across two specialized environments:

Mobile Application: Optimized for Faculty Panelists, temporary Guest Panelists, and IT Students, this interface prioritizes on-the-go accessibility. For evaluators, it functions as a real-time Evaluation Engine during live defenses. Guest Panelists authenticate via dynamically provisioned access codes, eliminating the friction of permanent account registration. For Students, the app serves as a secure portal for submitting peer-to-peer evaluations, viewing locked grades, and browsing the read-only digital vault. To protect intellectual property, the mobile vault strictly limits student viewing to authorized stage-specific deliverables while actively restricting access to full-length manuscripts.

Web Dashboard: Compiled via WebAssembly (Wasm) for near-native desktop performance, this centralized hub is designed for IT Administrators, PIT Leads, Capstone Advisers, and Repository Assistants. Administrators utilize the dashboard to configure academic periods and monitor Curriculum Analytics via the Decision Support System (DSS). To distribute administrative workloads, PIT Leads are granted delegated controls to configure rubrics, manage team formations, and route approved deliverables. Capstone Advisers utilize the dashboard to submit adviser-specific grades and upload final manuscripts, while Repository Assistants operate within a restricted, write-only portal to securely upload faculty-approved submissions into the archiving pipeline.

Application Logic Layer (Backend Server) Powered by Django and the Django REST Framework (DRF), the backend serves as the system's central processing hub, exposing secure APIs that facilitate communication between the clients and the database.

- **Automated Grading Engine:** Computes complex weighted percentage formulas for Capstone defenses while simultaneously aggregating raw point totals for PIT evaluations.
- **Grade Lock System:** Immediately secures scores and evaluation rubrics in the database upon submission, preventing any post-defense alterations and ensuring data integrity.
- **Validation and AI Pipeline:** Triggered upon document upload, this isolated two-stage NLP module processes deliverables prior to storage:
    - **The Gatekeeper (String Validation):** Utilizes strict Regular Expression (Regex) algorithms to verify that incoming file names comply with departmental conventions (e.g.,

YearLevel.Course.Semester). Non-compliant files are automatically rejected.

   - **The Classifier (Machine Learning):** Upon successful validation, the pipeline extracts document text. The TF-IDF algorithm vectorizes key terminology, and the Naive Bayes model calculates the probabilistic technological focus of the document (e.g., Machine Learning, Web Development). This autonomously routes the file to the correct thematic directory, feeding the DSS without human intervention.

Data Access Layer (Secure Repository and Database) This foundational layer manages state persistence and secure file storage, enforcing strict Role-Based Access Control (RBAC) at the database level.

- **Relational Database:** A PostgreSQL database securely stores encrypted user credentials, evaluation matrices, and the boolean flags governing the Grade Lock system.
- **Cloud Object Storage (Digital Vault):** Hosted via Amazon S3, this repository stores heavy media and PDF deliverables. To ensure ISO audit compliance and enforce physical library requirements, the vault applies Role-Based Visibility Filtering. While IT Administrators and Panelists are granted unrestricted access to complete manuscripts, general student access

is strictly limited to stage-specific deliverables (PIT documents, preliminary proposals, and summarized media). Furthermore, the storage layer dynamically applies watermarks to viewed documents to deter unauthorized reproduction.

3.4.1 Overview of the Proposed System

DefenSYS is a unified integration platform designed to modernize the evaluation and documentation workflows of the Department of Information Technology at the University of Science and Technology of Southern Philippines (USTP). Operating on a hybrid Client-Server architecture, the platform bridges a Web-Based administrative portal with a highly responsive cross-platform mobile application for real-time assessment and student interaction.

As illustrated in the system architecture diagram, DefenSYS is organized into five distinct, interconnected modules:

- **Administrative and Analytics Core:** Accessible via the web dashboard, this module serves as the system's administrative nerve center. It empowers IT Administrators to manage user accounts, configure academic terms, and define grading rubrics. Additionally, it houses the Decision Support System (DSS), which continuously mines repository data to generate empirical Curriculum Analytics that spotlight emerging technology trends.

- **Hybrid Evaluation and Grading Module:** Utilized by Faculty Members (Advisers, PIT Leads, Panelists) and temporary Guest Panelists, this cross-platform engine executes a dynamic, hierarchical grading workflow. For Capstone defenses, it automatically computes final grades based on strict weighted distributions: Panelists (50%), Advisers (30%), and anonymous Peer-to-Peer evaluations (20%). For Project Integration Technology (PIT) defenses, the module functions as a centralized tracker, accumulating raw evaluation scores for instructors.
- **Secure Student Dashboard & Mobile Vault:** This module provides students with a secure interface to monitor official team profiles and execute real-time peer evaluations. It integrates directly with the automated "Grade Lock" feature, which permanently secures scores in the database immediately upon submission. Furthermore, it provides read-only access to the digital vault, safely restricting general student viewing strictly to authorized, stage-specific deliverables to protect intellectual property.
- **Validation and Machine Learning Pipeline:** This module helps organize uploaded documents before they are stored in the repository. First, the Regex Gatekeeper checks whether the file follows the required departmental naming format. Files with incorrect names are rejected for correction. After validation, the system extracts text from the document and uses TF-IDF and Naive Bayes to classify it according to its main

technological theme. The classifier will use approximately 25 approved Capstone and PIT documents provided by the Department Chairman as its initial reference dataset. These documents will be labeled based on themes such as Web Development, Mobile Application, Machine Learning, Internet of Things, Information Systems, Data Analytics, and Cybersecurity. The labels will be verified by the Department Chairman or selected faculty members. Once classified, the file is routed to the proper repository category and protected using Role-Based Visibility Filters.

- **Centralized Data Management:** A unified PostgreSQL and Amazon S3 database infrastructure synchronizes all platforms and modules. Consequently, a grade entered via a mobile device during a live defense is instantaneously reflected on the Administrator's web dashboard, ensuring perfect real-time data consistency and eliminating operational fragmentation across the department.

3.4.2 System Objectives

General Objective

The general objective is to develop DefenSYS, a Cross-Platform Capstone and PIT Management System engineered to modernize and functionally unify the evaluation, secure archiving, and data analysis workflows within the Information Technology department.

Specific Objectives

The system development is guided by the following specific functional objectives:

1. To design a hybrid Web and Mobile Evaluation Module that supports weighted Capstone grading and raw PIT score aggregation.
2. To develop an automated file validation and archiving pipeline using Regex validation and repository routing.
3. To construct a secure digital repository with Role-Based Visibility Filtering and dynamic watermarking to protect student intellectual property.
4. To integrate a Machine Learning-based Curriculum Analytics module using TF-IDF and Naive Bayes to classify document themes and support DSS reporting.

3.4.3 System Tools Requirements

This section describes the specific hardware and software needed to build and run the DefenSYS platform. The main goals are to ensure compatibility across different systems, protect data, and enable effective machine learning.

Programming Language

Table 4. Programming Languages

Programming Language	**Version**
Dart	3.10.8
Python	3.14.3

Table 4 shows the programming languages the team plans to use in the system's development**.**

Hardware Requirements

Table 5. Hardware Minimum Requirements

Component	**Specifications**
Server Processor (CPU)	At least 4 cores and 8 threads
Server Memory (RAM)	At least 8 GB
Server Storage	Minimum 256 GB SSD
Mobile Processor	At least Snapdragon 400

Mobile Memory (RAM)	Minimum 3 GB RAM
Mobile Storage	Minimum 100 MB
Web Client Processor	Minimum Intel Core I3
Web Client Memory (RAM)	Minimum 4 GB RAM
Web Client Display	At least 720p
Web Client Connectivity	A reliable Internet connection

Table 5 shows the hardware requirements that the team intends to use in the development and design process.

Software Requirements

Table 6. Software Requirements

Application	Version
Visual Studio Code (IDE)	Latest Version
Android Studio	Otter 3 (2025.2.3)
PostgreSQL	Latest Stable
Amazon Cloud Server	S3
Frameworks	Version
Django	6.0.2
Flutter	3.38.9
Libraries	Version

Pdfplumber	Latest Stable
ReportLab	Latest Stable
Scikit-learn & NLTK	Latest Stable

Table 6 shows the software requirements for developing the system.

3.4.4 System Flow/Functions

The DefenSYS platform operates through a strict, sequential pipeline designed to ensure absolute data integrity from initial team formation through to the final archiving of documents. The workflow commences with an initialization phase, where IT Administrators and PIT Leads configure academic terms, establish rubrics, and register student teams via the centralized web dashboard. During the live evaluation phase, Faculty and Guest Panelists utilize the mobile Evaluation Engine to submit real-time scores, which the backend computes and immediately secures using a database-level "Grade Lock" to mathematically prevent unauthorized post-defense alterations. Finally, in the automated archiving phase, approved deliverables uploaded to the system pass through a strict Regex Gatekeeper for file-name validation. Validated files are then processed by a Machine Learning classifier (TF-IDF and Naive Bayes) that autonomously routes the documents into thematic directories while applying Role-Based Visibility controls to protect student intellectual property.

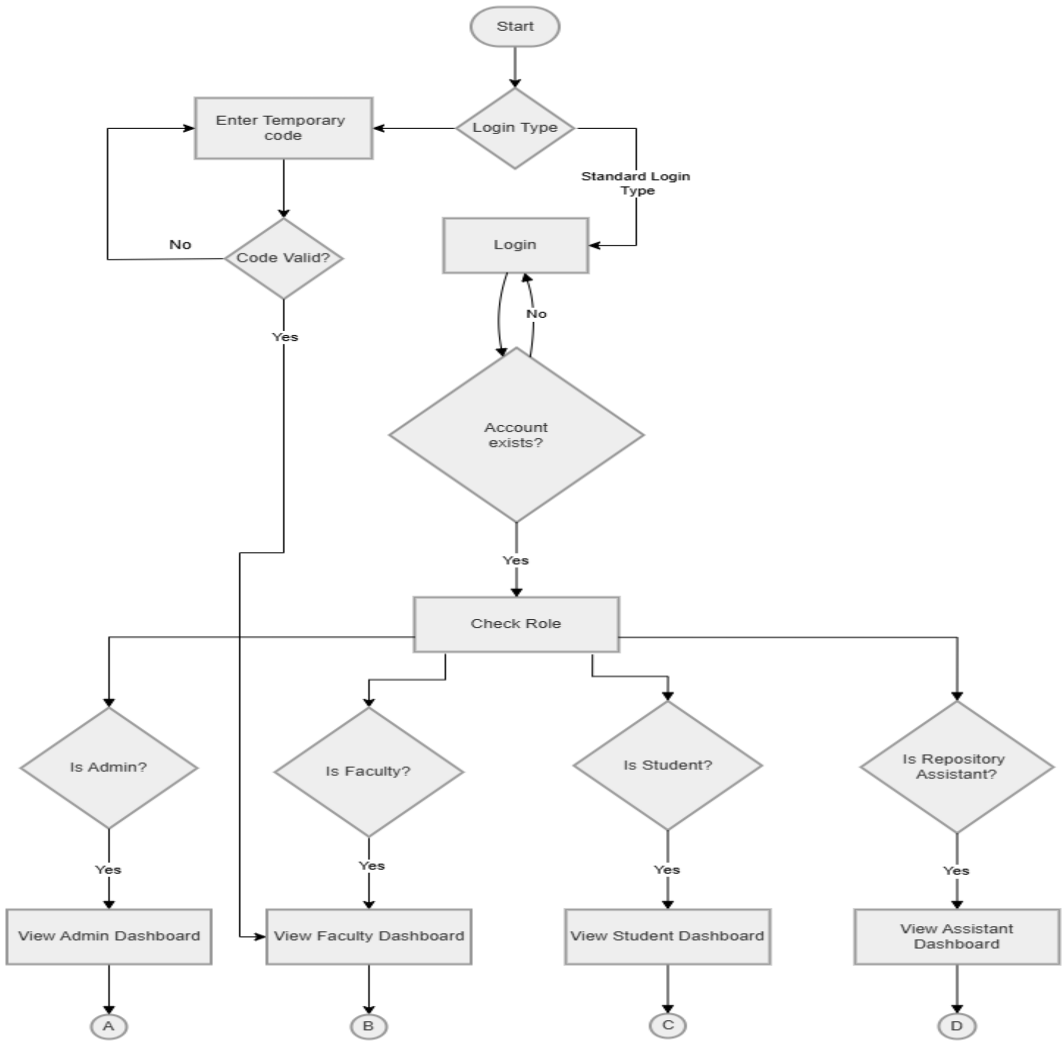

***Figure** 7. System Flow Chart (Log In)*

Figure 7 shows the System Flow Chart for the Log In process, detailing how the platform authenticates user credentials. Path A goes to admin flow, path B goes to faculty flow, path C goes to students flow, and path D goes to Repository flow.

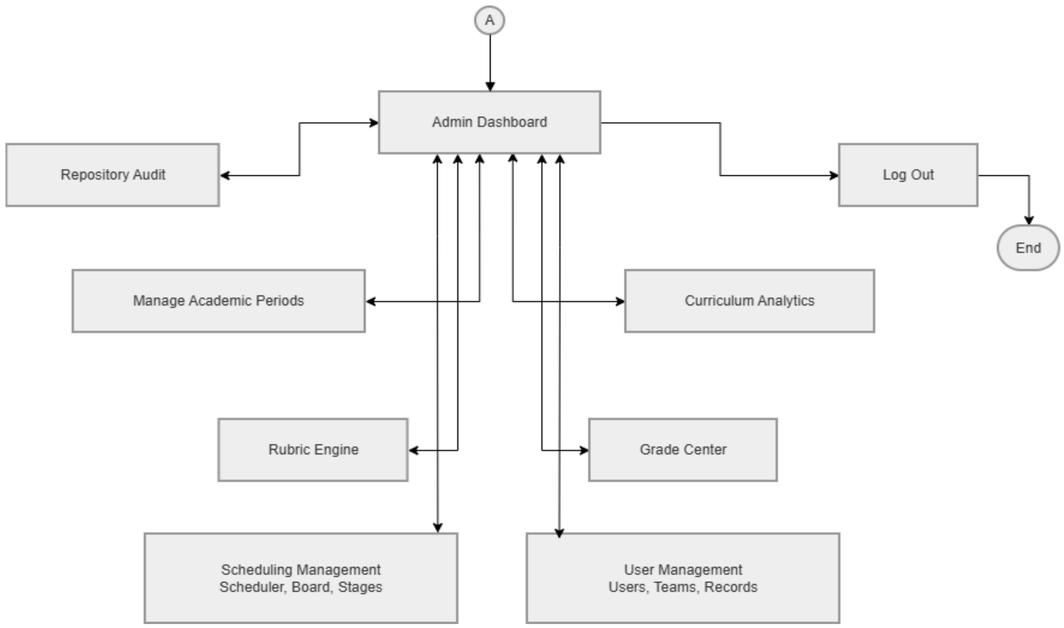

Figure 8. *System Flow Chart (Admin)*

Figure 8 shows the System Flow Chart for IT Administrators, mapping the workflow for managing academic periods, rubrics, and the repository audit.

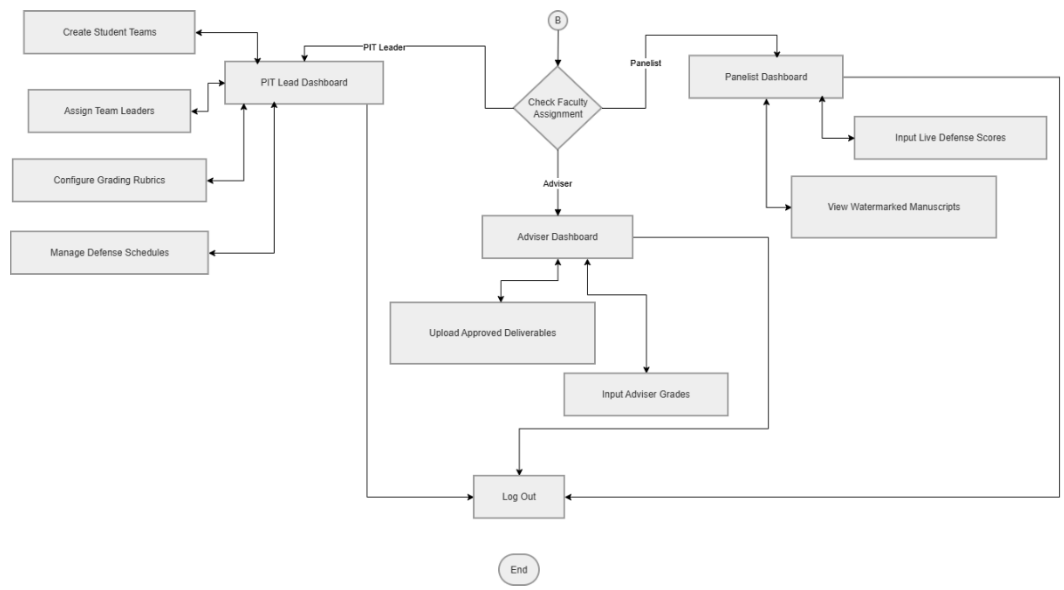

Figure 9. *System Flow Chart (Faculty)*

Figure 9 shows the System Flow Chart for Faculty (PIT Leads, Advisers, and Panelists), outlining the pathways for live grading and uploading approved deliverables.

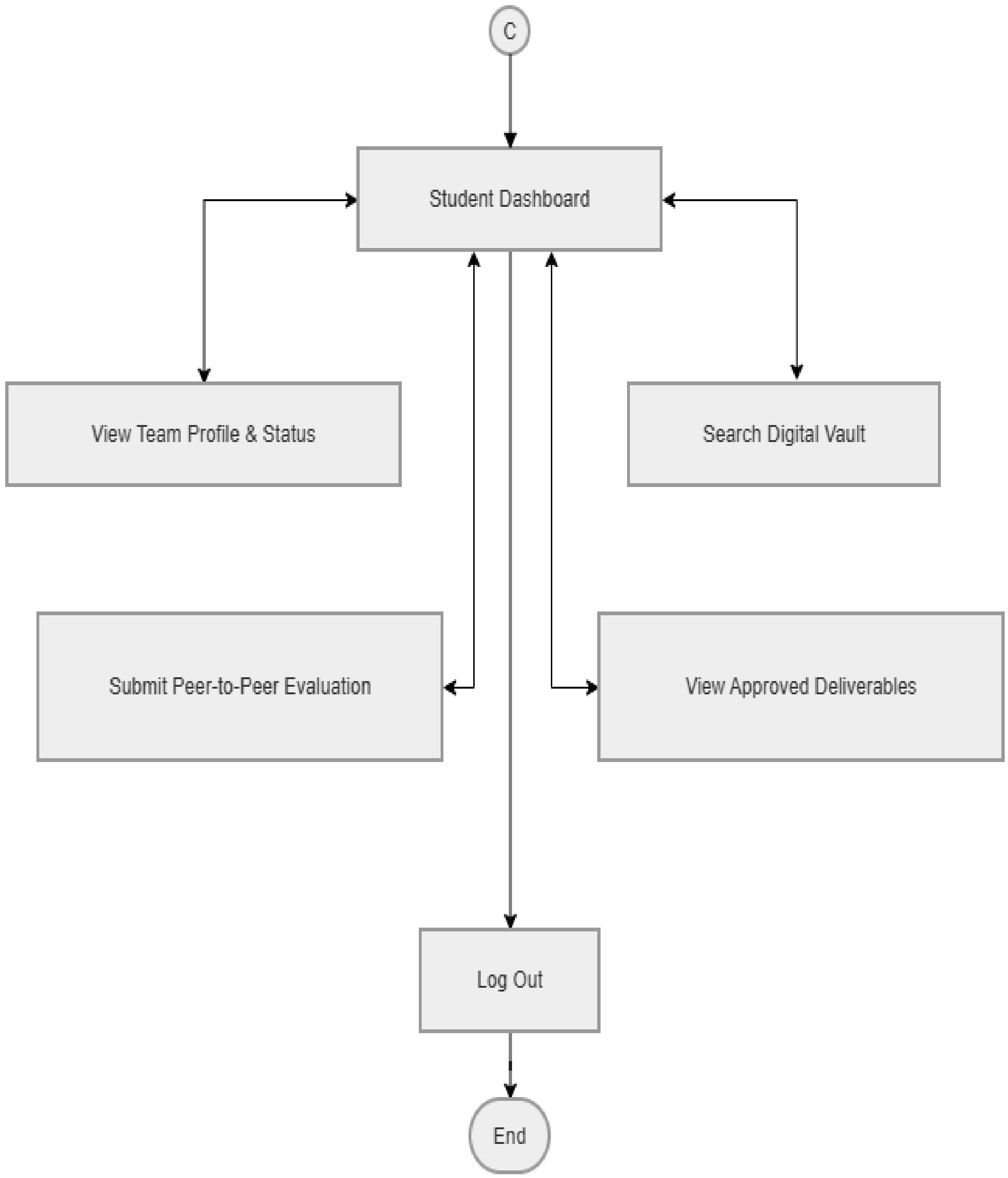

Figure 10. System Flow Chart (Student)

Figure 10 shows the System Flow Chart for Students, highlighting the steps for submitting peer evaluations and securely browsing the digital vault.

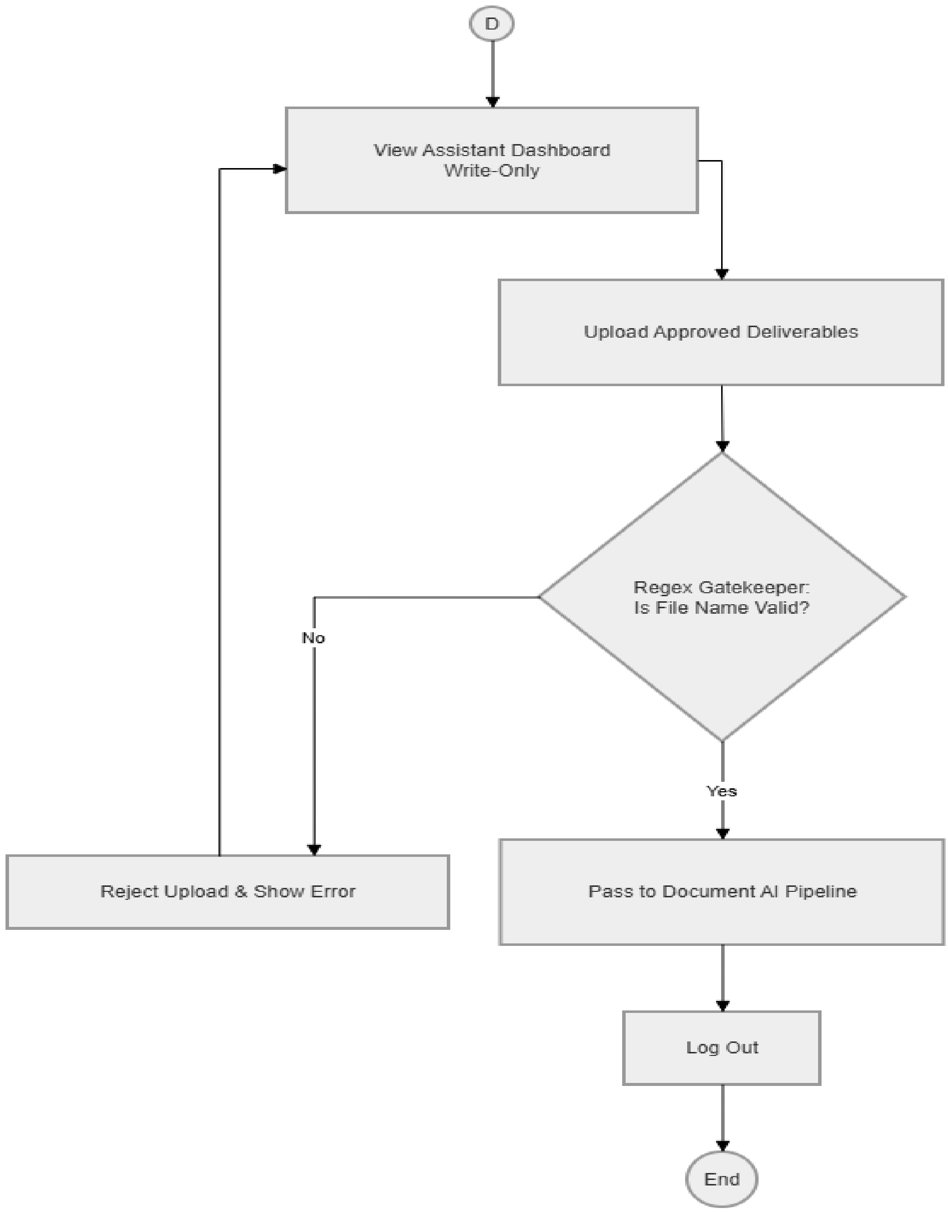

Figure 11. *System Flow Chart (Repository Assistant)*

Figure 11 shows the System Flow Chart for the Repository Assistant, showing the restricted process for uploading deliverables into the Regex Gatekeeper.

3.4.5 Scope and Limitations of the System

Scope of the System

The Cross-Platform Capstone and PIT Management System (DefenSYS) is strictly bounded to managing the evaluation, grading, and archiving workflows of the Project Integration Technology (PIT) and Capstone courses within the College of Information Technology. The system's operational capabilities encompass:

- Target Users: The system accommodates five distinct user roles governed by strict Role-Based Access Control (RBAC): IT Administrators, Faculty Members (PIT Leads, Capstone Advisers, and Panelists), IT Students (1st to 4th year), temporary Guest Panelists, and restricted-access Repository Assistants.
- Hybrid Evaluation and Grade Automation: The platform deploys a mobile application enabling evaluators to execute real-time grading during live defense sessions, alongside a student portal for synchronous peer-to-peer assessments. The Automated Grading Engine processes dual-grading workflows—computing complex weighted systemic grades for Capstone defenses and aggregating raw points for PIT evaluations. A database-level "Grade Lock" ensures all submitted scores are permanently secured and immutable.

- Automated Document Pipeline: The architecture utilizes a two-stage archival pipeline. First, a Regex Gatekeeper strictly enforces pre-configured departmental file-naming conventions prior to any upload. Second, a Machine Learning classifier (TF-IDF and Naive Bayes) extracts document text, mathematically categorizes the file by technological theme, and autonomously routes it to the correct departmental directory to feed the Decision Support System (DSS) without human intervention.
- Secure Archiving and Analytics: The system features a read-only digital vault equipped with dynamic watermarking and Role-Based Visibility Filtering. While complete manuscripts are securely archived for ISO audits and panelist evaluations, general student access is strictly limited to authorized stage-specific deliverables (PIT documents, preliminary proposals, and summarized media) to enforce the physical library mandate. Additionally, the DSS mines the repository to generate predictive Curriculum Analytics for administrative trend tracking.

System Limitations

To ensure focused development, structural integrity, and system stability, DefenSYS is subject to the following technical constraints:

- Network Dependency: Due to its cloud-hosted Client-Server architecture, both web and mobile environments require a stable, active broadband connection. The platform relies on real-time API communication for user authentication, live grading synchronization, and backend Machine Learning execution, thus precluding offline grading or local data caching.
- No Content Plagiarism Checking: While the AI pipeline categorizes documents by technological theme and extracts keywords, it does not evaluate qualitative academic merit. The system does not integrate with external plagiarism detectors (e.g., Turnitin), nor does it compile or check student source codes for syntax errors.
- Restricted Upload Privileges: To prevent database bloat and enforce strict quality assurance, students are entirely restricted from directly uploading deliverables to the digital vault. Only authorized Faculty Members and designated Repository Assistants possess the clearance to upload final, approved files through the Regex Gatekeeper.
- External Document Creation: DefenSYS functions exclusively as a management, evaluation, and repository platform; it does not feature an

integrated word processor, IDE, or code editor. Students must author their deliverables utilizing external software (e.g., Microsoft Word, Google Docs) prior to submitting the finalized files to their Adviser or Repository Assistant for systemic upload.

3.4.6 Calendar of Activities

Table 7. Calendar of Activities

DefenSYS (Calendar of Activities											
Activities	FEB	MAR	APRIL	MAY	JUNE	JULY	AUG	SEPT	OCT	NOV	DEC
Requirements Planning											
User Design & Prototyping											
Rapid Construction											
Iteration 1											
Iteration 2											
Iteration 3											
Cutover (Deployment)											

Table 7 shows the DefenSYS development timeline based on the iterative Rapid Application Development (RAD) methodology. The Gantt chart maps the overlapping execution of four core phases—Requirements Planning, User Design & Prototyping, Rapid Construction, and Cutover—ensuring continuous system testing and seamless deployment.

3.5 Conceptual Design and Implementation

The conceptual design of DefenSYS is explicitly engineered to eliminate the fragmented evaluation and archiving workflows within the Department of Information Technology at the University of Science and Technology of Southern Philippines (USTP). Fundamentally, the platform is structured as a secure, multi-tiered ecosystem that strategically decouples client-side data entry from resource-intensive backend processing. By decentralizing real-time evaluation to panelists' mobile devices while centralizing repository management and the Decision Support System (DSS) on an AI-powered web infrastructure, the system effectively neutralizes administrative bottlenecks. Furthermore, the core design guarantees absolute data integrity through the strict enforcement of a database-level Grade Lock mechanism.

To execute this conceptual architecture, the implementation phase strictly adheres to the iterative Rapid Application Development (RAD) methodology. Rather than utilizing a rigid, linear approach, the development team will rapidly construct and deploy functional prototypes of mission-critical modules—specifically the dual-grading Evaluation Engine and the Natural Language Processing (NLP) classifier for thematic directory routing. This iterative strategy mandates continuous, active feedback loops with IT Administrators and Faculty during user testing sessions. Consequently, developers can dynamically refine the cross-platform user interfaces, calibrate the automated rubric algorithms, and rigorously enforce the Role-Based Visibility Filtering protocols prior to the final system cutover.

3.5.1 Database Design

The database architecture for the Cross-Platform Capstone and PIT Management System (DefenSYS) will leverage a robust relational database model. To guarantee transactional data integrity, secure concurrent grading inputs, and ensure reliable archival indexing, the production environment will be engineered on PostgreSQL. The database schema will encompass 22 primary tables, systematically normalized and categorized into three interconnected operational modules:

1. Users and Academic Periods Module This foundational module will govern Role-Based Access Control (RBAC) and will tightly restrict system data visibility to designated academic terms.

- Core Tables: CustomUser, SchoolYear, Semester, YearLevel, StudentAcademicRecord, FacultyAssignment, and Team.
- Data Structure: The CustomUser table will act as the central entity, extending standard authentication to classify users into distinct RBAC profiles: IT Administrators, Faculty Members, IT Students, temporary Guest Panelists (utilizing auto-expiring access tokens with retained audit

logs), and write-only Repository Assistants. To preserve historical auditability, student and faculty records will be dynamically relation-mapped to the Semester and YearLevel tables. The Team table will strictly enforce business logic, mandating the assignment of exactly four students with matching academic credentials to a specific Capstone Adviser or PIT Lead.

2. Evaluation Engine Module Functioning as the central processing unit for all Capstone and PIT assessments, this module will manage the highly relational data required for multi-layered grading schemas.

- Core Tables: DefenseStage, DefenseSchedule, Rubric, Criterion, Evaluation, Rating, AdviserRating, and PeerEvaluation.
- Data Structure: The schema will utilize strict normalization to execute dual-grading computations. The Rubric table will map individual Criterion entries to their respective DefenseStage (e.g., Proposals vs. Final Hearings). This relational design will seamlessly accommodate both the complex weighted percentage formulas of Capstone defenses and the raw-point accumulations of PIT evaluations. During live defenses, the Evaluation table will act as the associative entity linking the Team, Rubric, and DefenseSchedule. As panelists, advisers, and peers submit scores (stored in Rating, AdviserRating, and PeerEvaluation respectively), the system will

automatically trigger a boolean "Grade Lock" flag, permanently securing the records against post-submission modification.

3. Digital Vault Module This module will manage the metadata and secure storage references for student intellectual property, actively enforcing the department's physical library mandate.

- Core Tables: Project, Document, and MediaAsset.
- Data Structure: Upon successful defense completion, a Team record will be converted and relationally linked to a definitive Project entry. The Document table will serve as the secure database registry for all uploaded deliverables. To satisfy physical library compliance, database queries will strictly apply Role-Based Visibility Filters. This design will grant Administrators and Faculty unrestricted query access to full manuscripts, while student queries will be securely restricted to approved, stage-specific records (PIT documentation, Capstone proposals, and summarized media). Crucially, the Document table will store validation metadata generated by the Regex Gatekeeper, alongside the analytical outputs from the Machine Learning pipeline—specifically the predicted technological_theme and the algorithm's confidence_score—to autonomously populate the Decision Support System (DSS).

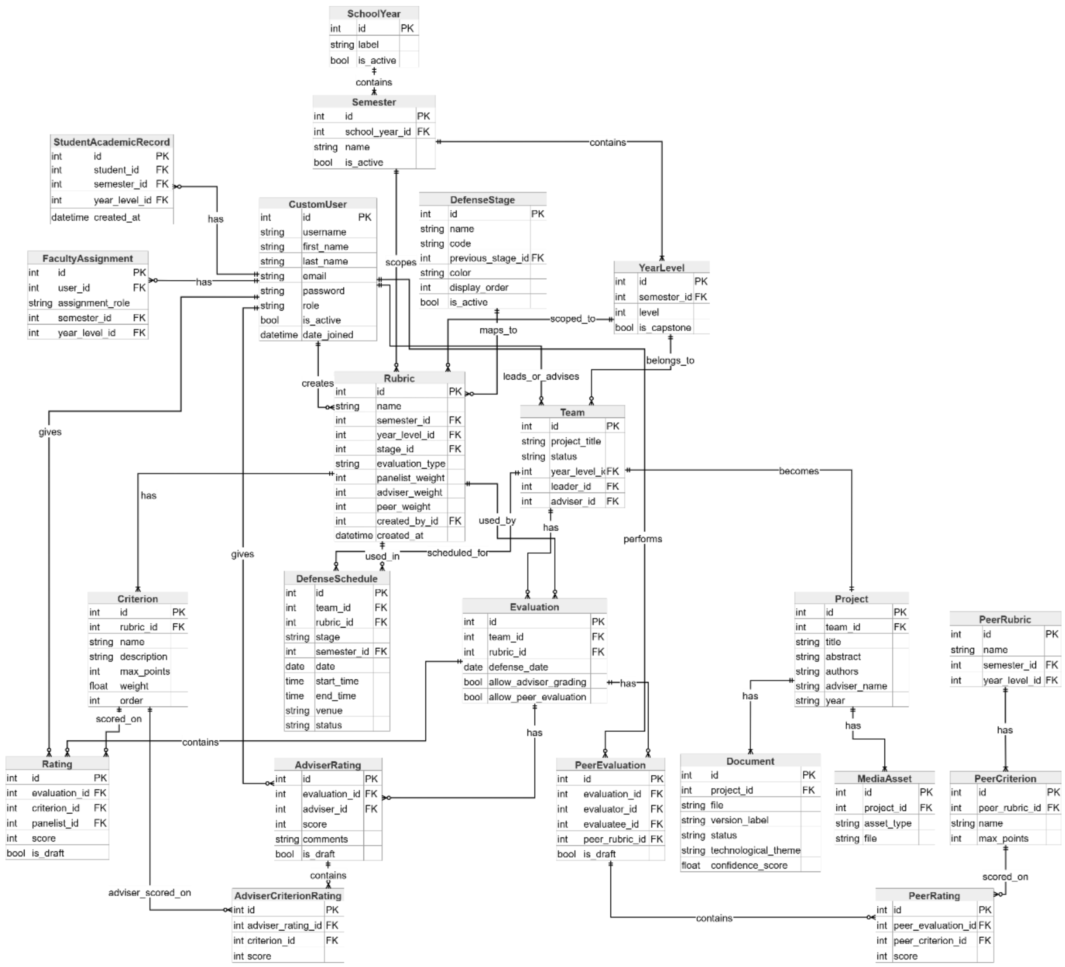

Figure 12. *Database Design*

Figure 12 shows the relational Database Design schema, detailing the strict entity relationships and primary key mappings across the authentication, evaluation, and digital vault modules.

3.5.2 System Design

The system design phase serves as the foundational architectural framework for the Cross-Platform Capstone and PIT Management System (DefenSYS), providing essential structural and behavioral models prior to active coding in the RAD lifecycle. By visually mapping the platform's multi-tiered architecture, this phase explicitly defines the operational boundaries and data routing mechanisms traversing from the client-side frontend interfaces to the centralized PostgreSQL database. These visualizations meticulously detail the interactions among discrete modules—such as the hybrid Evaluation Engine, Role-Based Access Control (RBAC) protocols, and the Automated Document Pipeline—thereby facilitating the seamless structural integration of critical features like the Grade Lock, Role-Based Visibility Filtering, and the Decision Support System (DSS).

Use Case Diagram

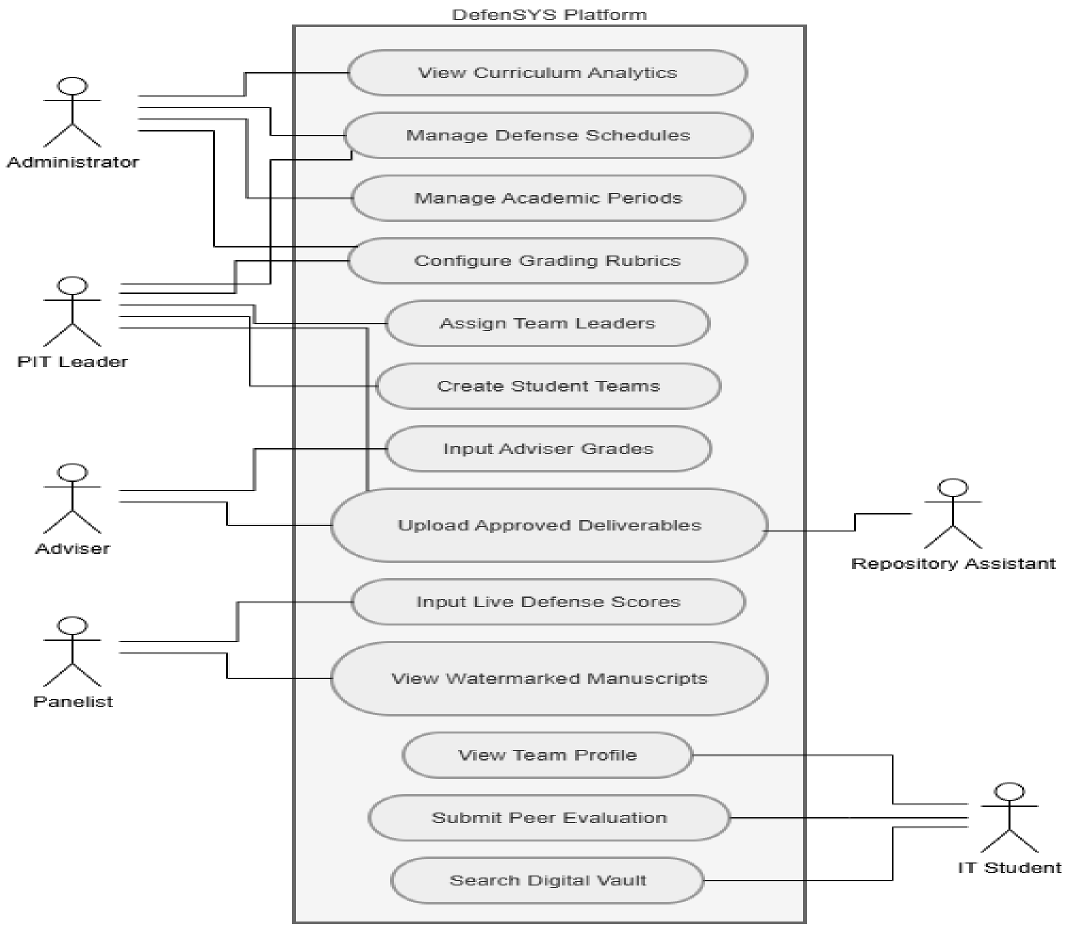

Figure 13. *Use Case Diagram*

Figure 13 shows the Use Case Diagram for the DefenSYS platform, visually mapping the behavioral interactions and specific Role-Based Access Control (RBAC) privileges across all five designated user groups. This explicitly defines the operational boundaries between administrative configurations, faculty evaluations, restricted repository uploads, and student access

Context Diagram

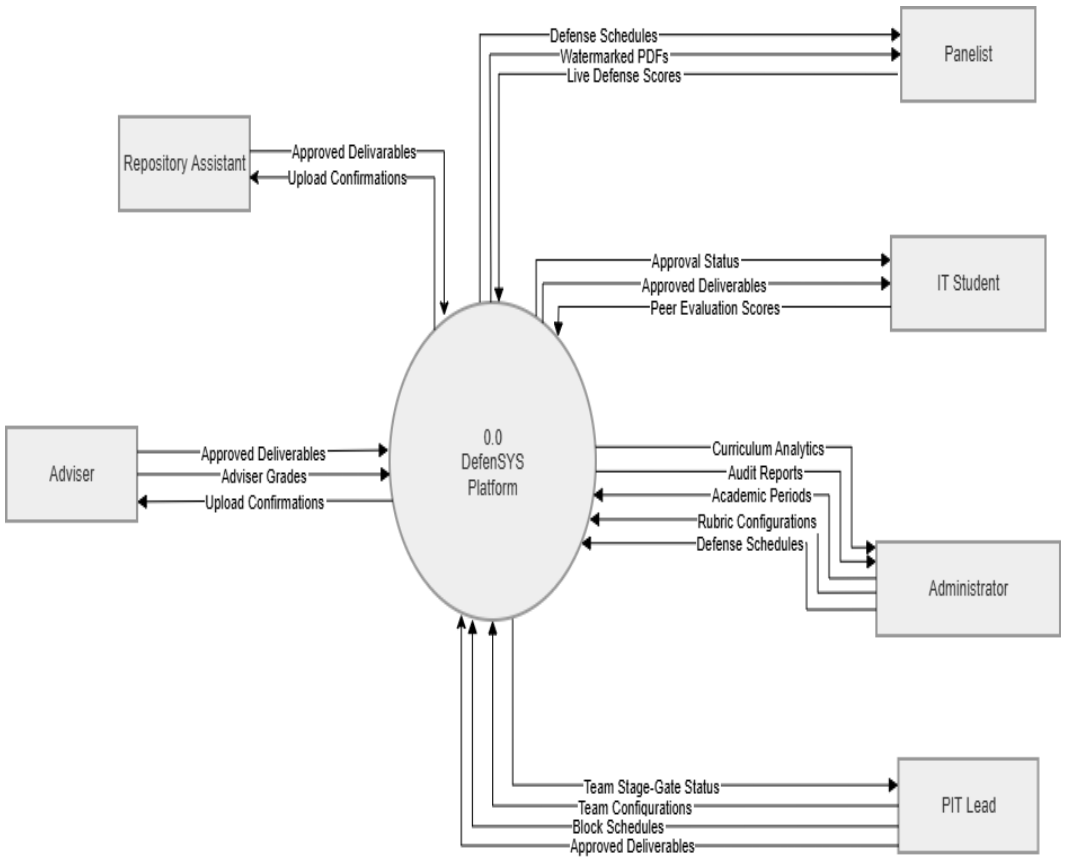

Figure 14. *Context Diagram*

Figure 14 shows the Context Diagram for the DefenSYS platform, illustrating the high-level data inputs and automated outputs exchanged between the centralized system and its five external user entities.

Data Flow Diagram

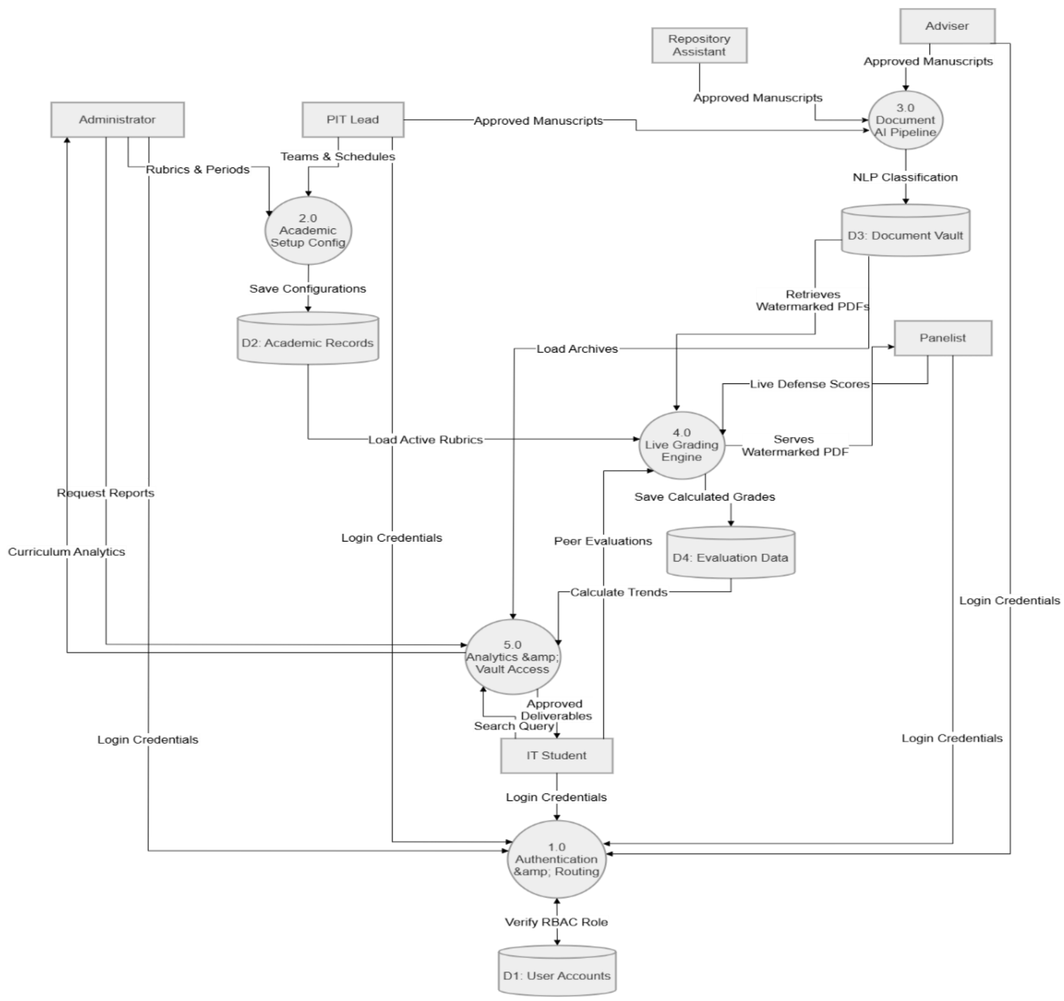

Figure 15. *Data Flow Diagram (DFD)*

Figure 15 shows the Data Flow Diagram (DFD) for the DefenSYS platform, visually mapping the internal backend transformations—from the mathematical aggregation of live evaluation scores to the AI-driven routing of repository deliverables.

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APPENDICES

Appendices A: Semi-Structured Interview Guide:

(Adopted and Adapted from Valacich & George [2020] for Business Process Analysis and Sommerville [2015] for Requirements Elicitation)

For the IT Department Head & Repository Admin

1. Can you describe the current "as-is" workflow for receiving, organizing, and archiving student Capstone and PIT manuscripts at the end of the academic period?
2. What are the primary technical or administrative bottlenecks you encounter regarding document standardization (e.g., file formats, naming conventions, version control) upon student submission?
3. Which specific phases of the current repository management workflow consume the highest amount of administrative time?
4. How does the current infrastructure enforce access control and data security to ensure that previous capstone source codes and manuscripts are protected from unauthorized extraction or plagiarism?
5. To what extent is historical repository data (e.g., programming languages, frameworks) currently utilized as a decision support tool to evaluate or update the IT curriculum?

For PIT Faculty Leads, Advisers, & Panelists

1. Describe the standard operating procedure you utilize when evaluating and grading a student group during a live defense session.
2. What are the most significant logistical frictions or usability issues you experience when utilizing current evaluation tools (e.g., paper rubrics, standalone spreadsheets) during a live defense?

3. What challenges exist in the current process of aggregating, consolidating, and computing final official grades from multiple evaluation sources (panelists, advisers, and peers)?
4. How does the current system allow you to track student group formations, manage composition changes, and monitor individual contributions throughout the semester?
5. What specific functional requirements or workflow automations would reduce your administrative load and allow you to focus more on the students' actual presentation?

For Senior IT Students (Capstone/PIT Enrollees)

1. Describe your user experience when attempting to retrieve or review past Capstone or PIT projects for literature reviews or format referencing.
2. What is your perception of the department's current ability to securely protect your original source code and completed manuscript from being copied by unauthorized users?
3. Describe the current workflow for conducting and submitting peer-to-peer evaluations. Do you feel the current method ensures data privacy and confidentiality?
4. What specific technical or procedural challenges do you face when submitting your final requirements to the department's repository?

Appendix B: Documentary Analysis Matrix

Document Category	Current Naming Convention	Storage Medium / Location	Security & Access Restrictions	Evaluation Weight (If Applicable)
PIT Deliverables				
Capstone Deliverables				
Capstone Proposals				
Full Manuscripts				
Source Codes				
Grading Rubrics				

Appendix C: Software Evaluation Questionnaire

ISO 25010 Criteria	**5**	**4**	**3**	**2**	**1**
I. Functional Suitability					
1. The mobile grading engine accurately computes the weighted percentages for Capstone and raw points for PIT.					
2. The Regex Gatekeeper successfully rejects files with incorrect naming conventions.					
3. The "Grade Lock" feature successfully prevents alterations after scores are submitted.					

II. Usability					
1. The mobile interface for live defense grading is easy to navigate for Faculty and Guest Panelists.					
2. The web dashboard provides a clear and organized view for managing academic periods and rubrics.					
3. Students can easily navigate the read-only digital vault to find authorized stage-specific deliverables.					
III. Reliability & Security					

1. The system strictly restricts general student access from viewing full-length manuscripts.					
2. The dynamic watermarking feature is consistently applied to documents in the digital vault.					
3. The AI Pipeline (TF-IDF & Naive Bayes) categorizes and routes documents without system crashing or data loss.					